

I Naive string matching alg

```
for i ← 0 to n-m
  j ← 0
  while j < m and p[j] = T[i+j] do
    j ← j+1
  if j == m return i
return -1
```

Tracing

T ← abcde, n=5

P ← cd, m=2

i ← 0 to 3

i=0 j=0

while j < m and p[j] = T[i+j]

0 < 2 and p[0] = T[0]

T and F = F

i=1 → 'i' is shifted by 1 in T

while 0 < 2 and p[0] = T[1]

T and F = F

✓ i=3 → 'i' is shifted by 1 in T

while 0 < 2 and p[0] = T[3]

C = C

T and T = T

j=1

~~i=4 → 'i' is shifted by 1 in T~~

while 1 < 2 and p[1] = T[4]
d = d

j=2

if j == m then return i

Pattern is found in i=3 position

II

KMP algorithm

① Computing π table

$m = P.length$

let $\pi[1..m]$ be a new array

$\pi(1) = 0$

$k = 0$

for $q = 2$ to m

{ while $k > 0$ and $P[k+1] \neq P[q]$

$k = \pi[k]$

if $P[k+1] = P[q]$

$k = k + 1$

$\pi[q] = k$

}

return π .

$P \leftarrow \text{abcabcd}, m = 7$

π

1	2	3	4	5	6	7
0	0	0	1	2	3	0

$k=0$ $q=2$

if $P[1] = P[2]$

$a \neq b$

$\pi[2] = 0$

$k=0$ $q=3$

if $P[1] = P[3]$

$a \neq c$

$\pi[3] = 0$

$k=0$ $q=4$

if $P[1] = P[4]$

$a = a$

$k=1$ $\pi[4] = 1$

$k=1$ $q=5$

if $P[2] = P[5]$, $k=2$, $\pi[5] = 2$
 $b = b$

$$\boxed{k=2} \quad \boxed{q=6}$$

if $p[3] = p[6]$, $k=3$, $\pi[6]=3$
 $c = c$

$$\boxed{k=3} \quad \boxed{q=7}$$

while $k > 0$ and $p[4] \neq p[7]$
 $a \neq q$

T and $T=T$

$$k = \pi[3] = 0$$

$$\pi[7] = 0$$

Thus the prefix table (π table is constructed.)

② Matching pattern in text (KMP-MATCHER)

$$n = T.length$$

$$m = P.length$$

// compute π table in π

$$q = 0$$

for $i = 1$ to n

{ while $q > 0$ and $p[q+1] \neq T[i]$

$$q = \pi[q]$$

if $p[q+1] == T[i]$

$$q = q + 1$$

if $q == m$

print "Pattern occurs with shift"
 $i - m$

$$q = \pi[q]$$

}

$T \leftarrow \text{abcabcabcabc}$, $n=10$
 $P \leftarrow \text{abcabcabc}$, $m=7$

$q=0$ $i=1$

if $p[i] == T[i]$
 $a == a$

$q=1$ $i=2$

if $p[i] == T[i]$
 $b == b$

$q=2$ $i=3$

if $p[i] == T[i]$
 $c == c$

$q=3$ $i=4$

if $p[i] == T[i]$
 $a == a$

$q=4$ $i=5$

if $p[i] == T[i]$
 $b == b$

$q=5$ $i=6$

if $p[i] == T[i]$
 $c == c$

$q=6$ $i=7$

while $q > 0$ and $p[i] \neq T[i]$
 $d \neq a$

T and $T = T$

$q = \pi(6) = 3$

if $p[i] == T[i]$
 $q=4$ $i=8$

if $p[i] == T[i]$

$q=5$ $i=9$

if $p[i] == T[i]$

$q=6$ $i=10$

if $p[i] == T[i]$
 $q=7$

$T \leftarrow \text{abcabcabcabc}$
 $P \leftarrow \text{abcabcabc}$

if $(q == m)$
 $7 == 7$

print 'pattern occurs with shift 3'

$q = \pi(7) = 0$

$T \leftarrow \text{abcabcabcabc}$
 $P \leftarrow \text{abcabcabc}$

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